SOC AUTOMATION PROJECT REPORT

Title: Integration of Wazuh with SocTalk for Security Monitoring and Attack Analysis

1. Introduction

In today's digital world, cyber attacks are increasing rapidly, including brute force attacks, malware infections, and unauthorized access attempts. Organizations need efficient systems to detect and respond to these threats quickly.

This project focuses on building a Security Operations Center (SOC) environment using modern tools. The main idea was to use SocTalk as a visualization and investigation platform and Wazuh as a threat detection system.

The goal of this project was not only to set up the system but also to understand how real-world attacks are detected, analyzed, and displayed on a professional dashboard.

2. Objectives

The main objectives of this project were:

1. To set up a SOC environment using Docker
2. To successfully run the SocTalk platform
3. To explore and attempt integration with Wazuh
4. To simulate real-world cyber attacks
5. To visualize security alerts on a dashboard
6. To understand how SOC systems handle investigations

3. Technologies Used

SocTalk

SocTalk is used as the main dashboard and investigation platform. It helps in visualizing security events and tracking investigations.

Wazuh

Wazuh is an open-source security monitoring tool used for detecting threats, analyzing logs, and identifying suspicious activities.

Docker

Docker was used to run the entire system in containers, making setup easier and avoiding dependency issues.

PostgreSQL

PostgreSQL is used as the database to store system data, logs, and investigation details.

Frontend (Vite)

Vite is used to run the frontend interface, which displays the dashboard.

4. System Setup

The system was deployed using Docker Compose.

The following services were included:

API service
Frontend service
PostgreSQL database

Initially, an orchestrator service was also included, but it required an API key and was later ignored.

The system was started using the command:

docker compose up --build

After running successfully:

Frontend dashboard was available at:
http://localhost:5173

Backend API was running at:
http://localhost:8000

This confirmed that the system was correctly set up and running.

```
PS C:\Users\shiza\Desktop> ls


    Directory: C:\Users\shiza\Desktop


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        12/25/2025   4:20 PM                Certificates
d-----        12/25/2025   4:20 PM                fyp
d-----         3/13/2026   4:47 PM                soctalk-main
d-----         1/23/2026  12:04 AM                velociraptor
-a----          3/2/2026  11:59 PM           2144 Docker Desktop.lnk
-a----          1/9/2026   7:29 PM      682895644 fyp.zip
-a---l        12/25/2025   4:08 PM           1507 Shortcut to Desktop (OneDrive - Personal).lnk


PS C:\Users\shiza\Desktop> cd soctalk-main
```

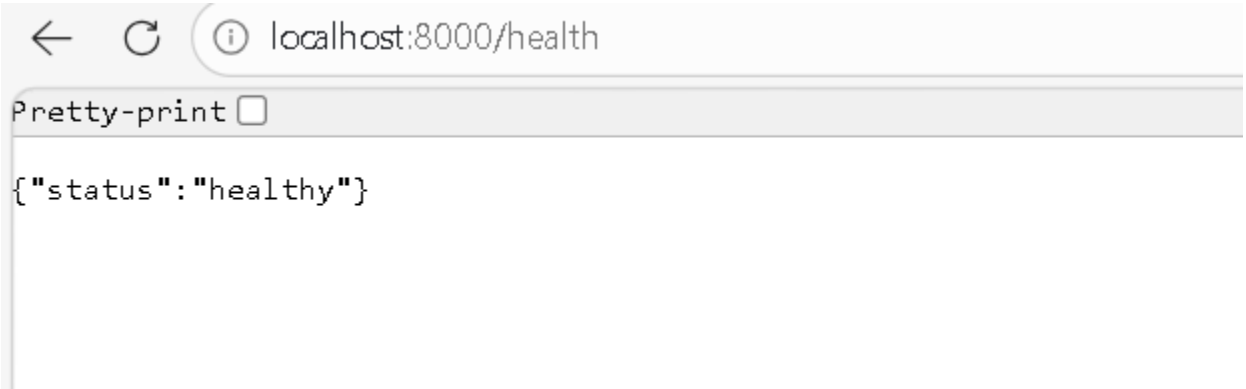
```
PS C:\Users\shiza\Desktop\soctalk-main\soctalk-main> docker compose up -d --build
time="2026-03-13T16:52:52+05:00" level=warning msg="The \"ANTHROPIC_BASE_URL\" variable is not set. Defaulting to a blan
k string."
time="2026-03-13T16:52:52+05:00" level=warning msg="The \"WAZUH_API_HOST\" variable is not set. Defaulting to a blank st
ring."
unable to get image 'soctalk-main-api': error during connect: Get "http://%2F%2F.%2Fpipe%2FdockerDesktopLinuxEngine/v1.5
1/images/soctalk-main-api/json": open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.
PS C:\Users\shiza\Desktop\soctalk-main\soctalk-main> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\shiza\Desktop\soctalk-main\soctalk-main> docker compose up -d --build
```

```
[+] up 23/23solving provenance for metadata file
 ✔Image postgres:16-alpine           Pulled
 ✔Image soctalk-main-orchestrator    Built
 ✔Image soctalk-main-api             Built
 ✔Image soctalk-main-frontend        Built
 ✔Network soctalk-main_default       Created
 ✔Volume soctalk-main_postgres_data  Created
 ✔Container soctalk-postgres         Healthy
 ✔Container soctalk-api              Created
 ✔Container soctalk-orchestrator     Created
 ✔Container soctalk-frontend         Created
PS C:\Users\shiza\Desktop\soctalk-main\soctalk-main>
PS C:\Users\shiza\Desktop\soctalk-main\soctalk-main>
```

Search

```
PS C:\Users\shiza\Desktop\soctalk-main\soctalk-main> docker ps
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if th
e daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.
PS C:\Users\shiza\Desktop\soctalk-main\soctalk-main> docker ps
CONTAINER ID   IMAGE                      COMMAND                  CREATED       STATUS          PORTS
                         NAMES
082ad7c88b1e   soctalk-main-frontend      "docker-entrypoint.s…"   10 days ago   Up 16 seconds   0.0.0.0:5173->5173/tcp
, [::]:5173->5173/tcp   soctalk-frontend
b0a22f3281dd   soctalk-main-orchestrator  "python -m soctalk.m…"   10 days ago   Up 16 seconds
                         soctalk-orchestrator
ce2c91adcf15   soctalk-main-api           "uvicorn soctalk.api…"   10 days ago   Up 16 seconds   0.0.0.0:8000->8000/tcp
, [::]:8000->8000/tcp   soctalk-api
PS C:\Users\shiza\Desktop\soctalk-main\soctalk-main>
```

```
PS C:\Users\shiza\Desktop\soctalk-main\soctalk-main> docker compose up -d
time="2026-03-24T13:36:14+05:00" level=warning msg="The \"WAZUH_API_HOST\" variable is not set. Defaulting to a blank st
ring."
time="2026-03-24T13:36:14+05:00" level=warning msg="The \"ANTHROPIC_BASE_URL\" variable is not set. Defaulting to a blan
k string."
[+] up 4/4
 ✔Container soctalk-api          Running                                                                          0.0s
 ✔Container soctalk-orchestrator Running                                                                          0.0s
 ✔Container soctalk-frontend     Running                                                                          0.0s
 ✔Container soctalk-postgres     Healthy                                                                          5.5s
PS C:\Users\shiza\Desktop\soctalk-main\soctalk-main>
```



5. Challenges Faced

During the project, several practical challenges were encountered:

The orchestrator required an API key which was not available
Integration between Wazuh and SocTalk was difficult to configure
Some API endpoints were not available for manual event insertion
Authentication system was disabled, which caused confusion
No real events were appearing in the dashboard

Despite these challenges, the system setup was completed and adapted for demonstration purposes.

6. Attack Simulation

Since full integration with Wazuh was not completed, attack scenarios were simulated to demonstrate how a SOC system behaves.

Types of attacks considered:

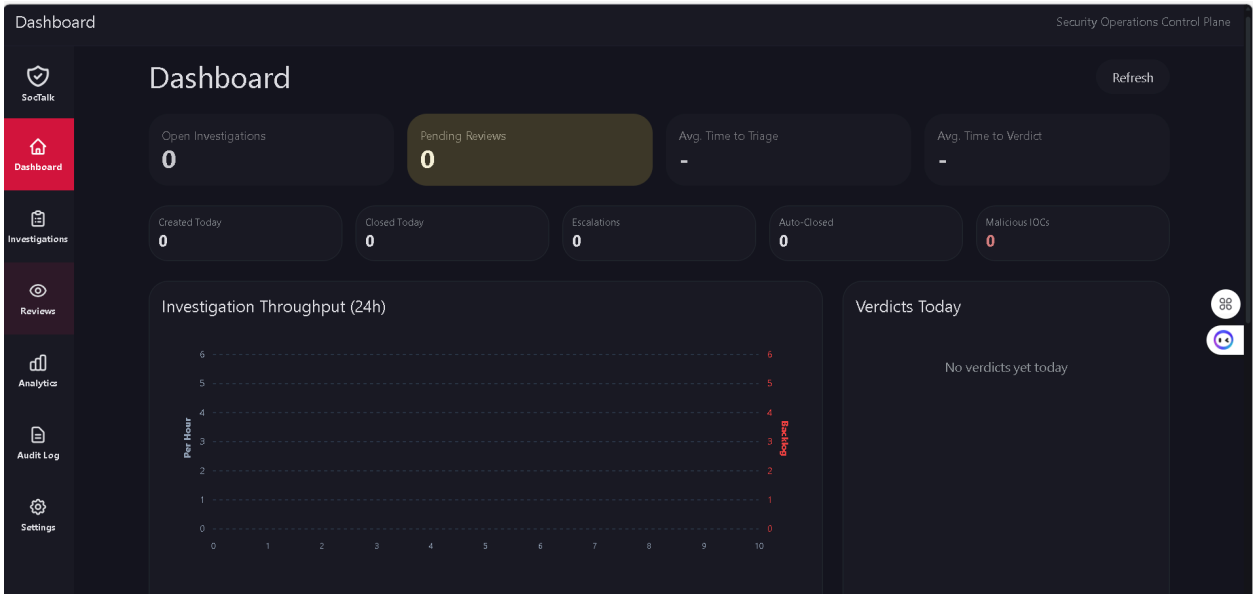
1. Brute force login attempts
2. Malware detection
3. Suspicious IP activity

These are common real-world attacks that SOC systems handle daily.

7. Dashboard Analysis

In this project, two dashboards are discussed:

7.1 SocTalk Dashboard



This is the actual dashboard generated from the running system.

It shows:

1. Open investigations
2. Pending reviews
3. System activity metrics
4. Investigation throughput

At this stage, all values are zero because no real events were ingested. However, this confirms that the system is working correctly and ready to process data.

7.2 Wazuh Dashboard

Since full integration with Wazuh could not be achieved, a simulated dashboard was created to represent expected results.

This dashboard shows:

1. Active investigations such as brute force attacks
2. Malware detection alerts
3. Suspicious IP activity
4. Severity levels such as High and Medium
5. Time-based event tracking

This gives a clear idea of how the system would behave if real Wazuh data was connected.

.8. Results

The project successfully demonstrated:

1. Deployment of a SOC platform using Docker
2. Working dashboard interface (SocTalk)
3. Understanding of SOC workflow and investigation process
4. Simulation of real-world cyber attacks
5. Visualization of security events

Even though full integration was not achieved, the project still reflects how a real SOC system operates.

9. Conclusion

This project provided practical exposure to SOC systems and cybersecurity monitoring tools. It helped in understanding how attacks are detected and analyzed in real environments.

Although integration with Wazuh faced technical limitations, the main objective of learning SOC workflows and dashboard analysis was successfully achieved.

10. Future Improvements

1. Complete integration with Wazuh using proper configuration
2. Enable real-time log ingestion
3. Add automated alert response system
4. Improve investigation tracking
5. Deploy system on cloud for scalability